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Topics to learn

1. Gradle project structure

- ✓ `src/main/java`, `resources`, `build.gradle`
 - ✓ Tasks, dependencies, lifecycle
- This is **build-time understanding**

2. Plain Java → Spring → Spring Boot

- ✓ Object creation (`new`)
- ✓ IoC / DI
- ✓ `@Component`, `@Configuration`
- ✓ Auto-configuration

This is **runtime architecture understanding**

3. Spring Environment

- ✓ `application.properties` / `yaml`
- ✓ Profiles
- ✓ Property resolution order

This is **externalized configuration understanding**

4. Spring Bean Lifecycle (VERY IMPORTANT)

When exactly is my object created, initialized, destroyed?

As part of this, you will learn

- Bean creation phases
- `@PostConstruct`
- `@PreDestroy`
- `InitializingBean`
- `BeanPostProcessor`
- Lazy vs eager beans

Because:

- Environment is read **before beans**
- Auto-configuration depends on lifecycle

5. Configuration styles (CORE SKILL)

Understand **3 ways of configuring Spring**:

1. Java Config (@Configuration)
2. Component scanning
3. Auto-configuration (Boot magic)

Focus on:

```
@ConditionalOnProperty  
@ConditionalOnClass  
@Profile
```

This connects **Environment** → **Bean creation**

6. Dependency Injection patterns (DESIGN LEVEL)

Learn **how beans should depend on each other**:

- ✓ Constructor injection (best practice)
- ✗ Field injection (why it's bad)
- ✓ Qualifiers
- ✓ Primary beans

This is where **clean architecture** starts.

7. Spring Boot auto-configuration (THE MAGIC)

This is the **most misunderstood topic**.

Learn:

- `spring.factories`
- `AutoConfiguration.imports`
- How Boot decides **what to create**
- Why adding a dependency changes behavior

This explains *why adding `spring-boot-starter-data-jpa` suddenly creates a `DataSource`*

8. Profiles + ConfigurationProperties (REAL WORLD)

You already touched `Environment`, now master:

- `@ConfigurationProperties`
- `@Validated`
- Profile-specific configs:
 - `application-dev.yml`
 - `application-prod.yml`
 -

This is **enterprise-grade configuration**.

9. Logging & Observability (ESSENTIAL)

Learn:

- SLF4J
- Logback
- Log levels per profile
- MDC
- Structured logging

Why?

Logs are your **first debugger in production**

10. Exception handling & Validation

Learn:

- `@ControllerAdvice`
- Global exception handling
- Bean validation (`@NotNull`, `@Size`)
- Custom exceptions

This separates **toy apps** from **production apps**.

11. Web layer (after foundations)

Only now move to:

- Spring MVC
- REST controllers
- Filters vs Interceptors
- Request lifecycle

Why later?

Web is just **one entry point** to Spring.

12. Data access (JPA/JDBC)

Learn:

- Transaction boundaries
- `@Transactional`
- Entity lifecycle
- Lazy vs eager fetching

13. Testing (VERY IMPORTANT)

Learn:

- Unit vs integration tests
- `@SpringBootTest`
- Slice tests
- Mocking beans

Plain Java

College class (simple POJO)

```
package com.testing.qe;

public class College {

    public void principalName() {
        System.out.println("Principal is Dr. Sharma");
    }
}
```

If you look at above Java code

- No annotations
- No framework
- Just a normal Java class

Main class (YOU manage everything)

```
package com.testing.qe;

public class TestApplication {

    public static void main(String[] args) {

        // YOU create the object
        College college = new College();

        // YOU call the method
        college.principalName();
    }
}
```

No Spring means:

Responsibility	Who does it now?
Object creation	YOU
Dependency management	YOU
Lifecycle control	YOU
Configuration	YOU
Scalability support	×
Loose coupling	×

Coupling

With out Spring : Tight coupling

```
public class College {  
    private Principal principal = new Principal(); // hard-coded  
    public void showPrincipal() {  
        principal.name();  
    }  
}
```

Problems:

- Cannot replace `Principal`
- Hard to test
- Code changes needed everywhere

With Spring : Loose coupling

```
@Component  
public class College {  
    @Autowired  
    private Principal principal;  
}
```

Spring:

- Creates `Principal`
- Injects it
- Allows easy replacement

Factory-style Java

Before Spring existed, people did this

```
public class ObjectFactory {  
  
    public static College getCollege() {  
        return new College();  
    }  
}
```

Usage:

```
College college = ObjectFactory.getCollege();
```

Spring is basically a **VERY powerful, configurable factory**

Why plain Java does NOT scale well

Imagine:

- 50 classes
- Each depends on 5 others

Without Spring:

- 250+ `new` statements
- Tight coupling
- Nightmare to maintain

With Spring:

- Zero `new`
- Configuration-based wiring

Interview

- ✓ Small utility programs
- ✓ Simple learning demos
- ✓ One-class applications

- ✗ Enterprise apps
- ✗ REST APIs
- ✗ Database-heavy systems

Spring framework

Example

```
package com.testing.qe;

import org.springframework.context.ApplicationContext;
import org.springframework.context.annotation.AnnotationConfigApplicationContext;

public class Test {
    public static void main(String[] args) {
        ApplicationContext context = new AnnotationConfigApplicationContext(CollegeConfig.class);
        College college = context.getBean("college", College.class);
        college.principalName();
    }
}
```

package

package com.testing.qe;

- This defines the **package (namespace)** of the class.
- It helps:
 - Organize code
 - Avoid class name conflicts
 - Enable **component scanning** in Spring

Import Statements

import org.springframework.context.ApplicationContext;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;

ApplicationContext	AnnotationConfigApplicationContext
• Central Spring IoC container • Responsible for: ◦ Creating objects (beans) ◦ Managing their lifecycle ◦ Injecting dependencies Think of it as Spring's factory + manager.	• A concrete implementation of <code>ApplicationContext</code> • Used when: ◦ Configuration is done using annotations ◦ No <code>applicationContext.xml</code> is used This class: • Reads <code>@Configuration</code> • Scans <code>@Component</code> • Handles <code>@Bean</code>, <code>@Autowired</code>, etc.

Class declaration

```
public class Test {
```

This is a **plain Java class**, not managed by Spring.

⚠️ Important:

- `Test` itself is **not a Spring bean**
- It just **boots the Spring container**

main method

```
public static void main(String[] args) {
```

- Java application entry point
- Starts Spring manually (not Spring Boot)

Creating Spring Container

```
ApplicationContext context = new AnnotationConfigApplicationContext(CollegeConfig.class);
```

What happens here step-by-step:

1. Spring container is created
2. `CollegeConfig.class` is loaded

```
@Configuration

@ComponentScan("com.testing.qe")

public class CollegeConfig {

}
```

3. Spring:

- Reads `@Configuration`
- Scans packages for:
 - `@Component`
 - `@Service`
 - `@Repository`
 - `@Controller`
- Creates beans

At this moment, all beans are created and stored in the container.

Getting a bean from Spring

```
College college = context.getBean("college", College.class);
```

What this means:

- "college" → **bean ID**
- `College.class` → expected type

Spring:

1. Looks for a bean named **college**
2. Confirms it is of type `College`
3. Returns the object

Usually defined as:

```
@Component
public class College {
    public void principalName() {
        System.out.println("Principal is Dr. Sharma");
    }
}
```

Bean name defaults to **class name with first letter lowercase**.

Calling business logic

```
college.principalName();
```

- Calls method on Spring-managed object
- Any dependencies inside `College` are:
 - Already injected
 - Already initialized

Execution flow

JVM starts

↓

main() method executes

↓

Spring container starts

↓

Configuration class loaded

↓

Beans created & wired

↓

getBean() fetches object

↓

Method executed

Spring Boot

Compare **your above Spring code** with **Spring Boot**, then we will cover **benefits** and **how to execute**.

Your CURRENT code (Plain Spring)

```
ApplicationContext context = new AnnotationConfigApplicationContext(CollegeConfig.class);
```

```
College college = context.getBean("college", College.class);
```

```
college.principalName();
```

What YOU do manually

- Create `ApplicationContext`
- Create `@Configuration` class
- Add `@ComponentScan`
- Manage dependencies
- Decide how the app starts

This is **pure Spring Framework**.

Above application using Spring Boot

Main Class

```
package com.testing.qe;
```

```
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;
```

```
@SpringBootApplication
public class TestApplication {

    public static void main(String[] args) {
        ApplicationContext context = SpringApplication.run(TestApplication.class, args);
        College college = context.getBean(College.class);
        college.principalName();
    }
}
```

No `AnnotationConfigApplicationContext`

No `CollegeConfig` needed

College class (almost same)

```
package com.testing.qe;

import org.springframework.stereotype.Component;

@Component
public class College {
    public void principalName() {
        System.out.println("Principal is Dr. Sharma");
    }
}
```

What changed?

Plain Spring	Spring Boot
Manual config class	✗ Not needed
Manual component scan	✗ Auto
XML / Java config	✗ Optional
Embedded server	✗
Boilerplate code	✗

What `@SpringBootApplication` really doing?

`@SpringBootApplication`

This is **3 annotations combined**

`@Configuration`

`@EnableAutoConfiguration`

`@ComponentScan`

Meaning:

- `@Configuration` → Java config
- `@ComponentScan` → scans current package & subpackages
- `@EnableAutoConfiguration` → magic ✨ (auto setup)

BENEFITS of Spring Boot

✓ 1. No configuration headache

- No XML
- No manual bean wiring
- No server setup

✓ 2. Auto-configuration (HUGE)

Example:

- Add `spring-boot-starter-web`
- BOOM ☐
 - Tomcat configured
 - DispatcherServlet created
 - JSON mapper ready

You didn't write **anything**.

✓ 3. Embedded server

No need to install:

- Tomcat
- JBoss
- WebLogic

Spring Boot ships with **embedded Tomcat**.

✓ 4. Production-ready

Spring Boot gives:

- Health checks
- Metrics
- Logging
- Externalized config

How to EXECUTE Spring Boot app

Option 1: From IDE (IntelliJ / Eclipse)

- Right-click on `TestApplication`
- Click **Run**

✓ Most common

Option 2: Maven command

```
mvn spring-boot:run
```

Option 3: Run JAR (production way)

```
mvn clean package java -jar target/test-0.0.1-SNAPSHOT.jar
```

Project structure (Spring Boot way)

```
src/main/java
├── com.testing.qe
│   ├── TestApplication.java <-- main class
│   └── College.java
```

Main class must be at top package level

(or component scanning will fail)

Interview

Spring Boot = Spring + Auto Configuration + Embedded Server + Zero Boilerplate

When should YOU use Spring Boot?

- ✓ REST APIs
- ✓ Microservices
- ✓ Real-world projects
- ✓ Faster development

Environment in Spring

`org.springframework.core.env.Environment` is **one of the most important abstractions in Spring**, but it's often used without really understanding *why*.

What Environment actually is?

In **Spring Framework**, `Environment` is an **interface** that represents:

The current runtime environment of your application

It gives you **read-only access** to:

- Configuration **properties**
- Active **profiles**
- Default profiles

Why Spring needs `Environment`

Modern applications don't hardcode values like:

- Database URL
- Username/password
- Port numbers
- Feature flags

Instead, these values change by **environment**:

- dev
- test
- qa
- prod

`Environment` lets Spring **centralize and resolve** all of this configuration.

What can you do with `Environment`?

1. Read configuration properties

```
@Autowired
private Environment environment;

String url = environment.getProperty("spring.datasource.url");
```

This value could come from:

- `application.properties`
- `application.yml`
- OS environment variables
- JVM system properties
- Command-line arguments

You **don't** care where it came from — Spring resolves it.

2. Check which profile is active

```
if (environment.acceptsProfiles("prod")) {
    // production-only logic
}
```

Or:

```
String[] profiles = environment.getActiveProfiles();
```

This is extremely useful for:

- Logging
- Conditional behavior
- Feature toggles

3. Provide default values safely

```
String timeout =
    environment.getProperty("app.timeout", "30");
```

If `app.timeout` is missing → "30" is used.

No `NullPointerException`.

Why not just use `@Value`?

You *can* do this:

```
@Value("${server.port}")
private int port;
```

But `Environment` is better when:

Use case	Why <code>Environment</code> is better
Dynamic access	Can read properties anytime
Conditional logic	Profiles + properties together
Optional properties	Built-in defaults
Libraries/frameworks	Cannot rely on <code>@Value</code>

Typical real-world use cases

✓ Database switching

```
if (environment.acceptsProfiles("dev")) { // H2 database } else { // Oracle /
MySQL }
```

✓ Feature flags

```
if (environment.getProperty("feature.newUI", Boolean.class, false)) {
enableNewUI(); }
```

✓ Logging config

```
log.info("Active profile: {}", Arrays.toString(environment.getActiveProfiles()));
```

How Spring populates `Environment`

Spring collects properties from **PropertySources**, in this order:

1. Command-line args
2. JVM system properties
3. OS environment variables
4. `application.properties` / `yml`
5. `@PropertySource`

You can inspect them:

```
ConfigurableEnvironment env = (ConfigurableEnvironment) environment;  
env.getPropertySources().forEach(ps -> System.out.println(ps.getName()));
```

Spring Boot vs Spring Framework

In **Spring Boot**:

- Environment is **auto-configured**
- `application.properties` is loaded automatically
- Profiles are first-class citizens

In plain Spring:

- You must define property sources manually

But the **Environment API is identical**.

When SHOULD you use `Environment`?

- ✓ Framework / infrastructure code
- ✓ Conditional configuration
- ✓ Reading optional properties
- ✓ Profile-based behaviour

When you should NOT

- ✗ Simple POJOs
- ✗ Business logic classes
- ✗ Where `@ConfigurationProperties` fits better

High level comparision

Aspect	Environment	@Value	@ConfigurationProperties 
Purpose	Programmatic access to config	Inject single value	Bind structured config
Style	Imperative (code-driven)	Declarative (annotation)	Declarative + type-safe
Type safety	✗ No	⚠ Partial	✓ Yes
Supports defaults	✓ Yes	⚠ Limited	✓ Yes
Best for	Framework / infra logic	Simple values	Application config
Testability	⚠ Medium	✗ Poor	✓ Excellent
Validation	✗ No	✗ No	✓ Yes (<code>@Validated</code>)

Hand on – Exercise to do

Create Project structure

```
demo-env/  
├─ build.gradle  
└─ src/  
    └─ main/  
        ├─ java/  
        │   └─ com/example/envdemo/  
        │       ├─ EnvDemoApplication.java  
        │       └─ EnvPrinter.java  
        └─ resources/  
            ├─ application.properties  
            ├─ application-dev.yml  
            └─ application-prod.yml
```

Gradle build file

```
plugins {  
    id 'java'  
    id 'org.springframework.boot' version '3.2.1'  
    id 'io.spring.dependency-management' version '1.1.4'  
}  
  
group = 'com.example'  
version = '0.0.1-SNAPSHOT'  
  
java {  
    toolchain {  
        languageVersion = JavaLanguageVersion.of(17)  
    }  
}  
  
dependencies {  
    implementation 'org.springframework.boot:spring-boot-starter'  
}
```

Note: This uses **Spring Boot**, which auto-wires the `Environment`

application.properties (BASE configuration)

```
app.name=Env Demo App
app.timeout=30
app.source=application.properties
```

application-dev.yml (DEV profile)

```
app:
  timeout: 10
  source: application-dev.yml
```

Note: Overrides only when **dev** profile is active.

application-prod.yml (PROD profile)

```
app:
  timeout: 60
  source: application-prod.yml
```

Note: Overrides only when **prod** profile is active.

Main application class

```
package com.example.envdemo;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class EnvDemoApplication {

    public static void main(String[] args) {
        SpringApplication.run(EnvDemoApplication.class, args);
    }

}
```

Environment reader bean (MOST IMPORTANT PART)

```
package com.example.envdemo;

import jakarta.annotation.PostConstruct;
import org.springframework.core.env.Environment;
import org.springframework.stereotype.Component;

import java.util.Arrays;

@Component
public class EnvPrinter {

    private final Environment environment;

    public EnvPrinter(Environment environment) {
        this.environment = environment;
    }

    @PostConstruct
    public void print() {
        System.out.println("---- ENVIRONMENT DEMO ----");

        System.out.println("Active profiles: "
            + Arrays.toString(environment.getActiveProfiles()));

        System.out.println("app.name = "
            + environment.getProperty("app.name"));

        System.out.println("app.timeout = "
            + environment.getProperty("app.timeout"));

        System.out.println("app.source = "
            + environment.getProperty("app.source"));

        System.out.println("-----");
    }
}
```

Note: This bean runs **after the Environment is prepared**, but **before the app is fully ready**.

Run scenarios (THIS IS WHERE YOU LEARN)

Case 1: No profile (default)

bash

 Copy code

```
./gradlew bootRun
```

Output:

yaml

 Copy code

```
Active profiles: []
app.name = Env Demo App
app.timeout = 30
app.source = application.properties
```

✓ Only `application.properties` used

Case 2: DEV profile

bash

 Copy code

```
./gradlew bootRun --args='--spring.profiles.active=dev'
```

Output:

yaml

 Copy code

```
Active profiles: [dev]
app.name = Env Demo App
app.timeout = 10
app.source = application-dev.yml
```

✓ `application-dev.yml` overrides base

Case 3: PROD profile

bash

 Copy code

```
./gradlew bootRun --args='--spring.profiles.active=prod'
```

Output:

r

 Copy code

```
Active profiles: [prod]
app.name = Env Demo App
app.timeout = 60
app.source = application-prod.yml
```

Case 4: Command-line override (HIGHEST PRIORITY)

bash

 Copy code

```
./gradlew bootRun --args='--spring.profiles.active=dev --app.timeout=99'
```

Output:

yaml

 Copy code

```
Active profiles: [dev]
app.timeout = 99
```

✓ Command-line wins over everything

Property resolution order (MEMORIZE THIS)

From **highest** → **lowest** priority:

1. **Command-line arguments**
2. JVM system properties (`-D`)
3. OS environment variables
4. `application-{profile}.yml`
5. `application.properties`
6. Default values in code

That's why `--app.timeout=99` wins.

REST API - Application

Instead of calling:

```
college.principalName();
```

We will expose it as a **REST endpoint**:

```
GET http://localhost:8080/college/principal
```

and get a response like:

```
Principal is Dr. Sharma
```

Create a Spring Boot project

Use: IntelliJ / Eclipse **OR**

Required dependency

Add only **Spring Web**

(Maven dependency)

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-web</artifactId>
</dependency>
```

This is powered by **Spring Boot**, built on **Spring Framework**.

Project structure (IMPORTANT)

```
src/main/java
├── com.testing.qe
│   ├── TestApplication.java ← main class
│   ├── College.java         ← business logic
│   └── CollegeController.java ← REST API
```

Main class must be at **top package level**.

Main Application class

```
package com.testing.qe;
```

```
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
```

```
@SpringBootApplication
public class TestApplication {

    public static void main(String[] args) {
        SpringApplication.run(TestApplication.class, args);
    }
}
```

✓ No `ApplicationContext`

✓ No config class

✓ No manual wiring

Business logic (same as before)

```
College.java
```

```
package com.testing.qe;
```

```
import org.springframework.stereotype.Component;
```

```
@Component
public class College {

    public String principalName() {
        return "Principal is Dr. Sharma";
    }
}
```

✓ Spring manages this bean

✓ No `new College()` anywhere

REST Controller (NEW)

CollegeController.java

```
package com.testing.qe;
```

```
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
```

```
@RestController
public class CollegeController {

    @Autowired
    private College college;

    @GetMapping("/college/principal")
    public String getPrincipalName() {
        return college.principalName();
    }
}
```

What annotations mean (VERY IMPORTANT)

Annotation	Purpose
@RestController	Marks class as REST API
@GetMapping	Maps HTTP GET request
@Autowired	Injects <code>College</code> bean
@Component	Makes class a Spring bean
@SpringBootApplication	Auto config + scan

How execution works (FLOW)

Browser / Postman

|
| GET /college/principal
v

CollegeController

|
v

College (business logic)

|
v

Response returned

How to RUN the application

✓ Option 1: From IDE

- Right-click `TestApplication`
- Click **Run**

✓ Option 2: Maven

```
mvn spring-boot:run
```

✓ Option 3: JAR (production style)

```
mvn clean package java -jar target/*.jar
```

How to TEST the REST API

In browser

<http://localhost:8080/college/principal>

In curl

```
curl http://localhost:8080/college/principal
```

Why REST API is better than `main()` call

Old way

Console output

Local JVM

Not reusable

Hard to scale

REST way

HTTP response

Network accessible

Used by UI / Mobile

Cloud-ready

Interview one-liner

Spring Boot REST API exposes business logic using HTTP endpoints, making applications scalable, reusable, and platform-independent.

What next

- Convert response to **JSON**
- Add **POST API**
- Add **DTO**
- Connect to **Oracle DB**
- Explain **@RestController** vs **@Controller**